AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

First Review – B.Tech 7th Semester Project

Drought-Aware Crop Productivity Decision - Support System

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Motivation & Problem Statement

Motivation

5 Maharashtra scarcity-zone districts face recurring drought and dry spells.
Official Contingency Plans exist — but sit unused as static PDFs, read by hand.

Problem

No system links weather, yield, human impact, and official guidance together.
Drought is judged informally, not against each district's own 23-year history.
Matching a condition to its action means searching a 50+ page PDF by hand.

SCOPE — Explainable ML + knowledge-graph-grounded LLM · 5 districts · Kharif & Rabi seasons

1 · Classify Condition

Detect drought/dry-spell against each district's own history — not generic thresholds.

2 · Predict & Explain

SHAP-explainable yield forecast, translated into human impact via Census data.

3 · Ground Every Action

Knowledge graph + LLM retrieve official recommendations — never invent them.

Project Progress Since Zeroth Review

Data Preprocessing

Cleaned, standardized, and merged the district weather, crop-yield, and demographic datasets into one unified modeling table.

Exploratory Data Analysis

Analyzed trends, distributions, and correlations across all datasets to validate assumptions before modeling.

Model Comparison

Trained five candidate forecasting models and benchmarked their performance to select the best-fit approach.

	Model	Type	Purpose
1	**SARIMA**	Statistical	Baseline for temporal rainfall patterns and seasonality
2	**XGBoost**	Classical ML	Forecast rainfall using lag, rolling-window, and weather features
3	**LSTM**	Deep Learning	Learn sequential/temporal dependencies in historical weather
4	**Temporal Fusion Transformer (TFT)**	Transformer	Multivariate time-series forecasting with temporal attention
5	**Hurdle Model**	Two-stage model	First predicts **rain/no-rain**, then predicts **rainfall amount if rain occurs**

Why These 5 Districts

District	Region
Ahmednagar	Western Maharashtra – rain-shadow belt
Beed	Marathwada
Latur	Marathwada
Osmanabad	Marathwada
Solapur	Western Maharashtra – rain-shadow belt

Recurring Drought Exposure

All 5 districts lie in Maharashtra's traditional scarcity zone — the Marathwada plateau and the semi-arid rain-shadow belt — with a long history of failed or delayed monsoons.

Consistent Multi-Source Data

Each district offers 26 years of daily weather data (2000–2025), 23 years of crop records (2000-01 to 2022-23), and Census 2011 demographics — a uniform base for modeling.

Official Contingency Plans Exist

ICAR-format District Agriculture Contingency Plans already exist for exactly these 5 districts — the knowledge base the project's recommendation layer is grounded in.

Literature Review & Research Gap

S.No	Paper title and year	Key contributions	Limitations
1	Forecasting rainfall events based on zenith wet delay time series utilizing eXtreme gradient boosting (XGBoost)	Uses XGBoost with GNSS-derived Zenith Wet Delay (ZWD) features to predict rainfall occurrence one hour ahead, achieving reliable rainfall-event forecasting across multiple stations.	Short forecasting horizon (1 hour ahead) and requires GNSS ZWD data; evaluated mainly in the eastern USA, limiting direct generalization to other regions and weather regimes.
2	Forecasting temperature and rainfall using deep learning for the challenging climates of Northern India (2025)	Compares RNN and LSTM with single-variable and multi-variable setups on 24 years of daily IMD data from 3 northern Indian stations.	Only forecasts one day ahead from 3 variables, with no attention or added features.
3	Temporal Fusion Transformers for Interpretable Multi-Horizon Time Series Forecasting (2021)	Attention-based multi-horizon forecasting model. Provides interpretable variable selection and temporal attention.	Complex architecture with high computational cost. Requires substantial data and tuning.
4	Time Series Forecasting of Monthly Rainfall using Seasonal Auto Regressive Integrated Moving Average with Exogenous Variables (SARIMAX) Model (2024)	Forecasts monthly rainfall in Phaltan taluka, Maharashtra configurations — (1,0,1)(0,1,0,12) and (1,1,1)(1,1,1,12) — and compares them directly on R^2 and RMSE, The better model achieved RMSE of 54.54 and R^2 of 0.91, while the second scored RMSE 71.12 and R^2 0.8	Single-station study (one taluka, one rain gauge) — it doesn't test whether the same order choice generalizes across multiple districts with different rainfall regimes

Weather Data — Patterns & Variability

Daily weather records (2000–2025) across all 5 districts — rainfall, temperature, and dry-spell patterns

District	Mean Annual Rainfall (mm)	Monsoon Share (Jun–Sep)	Avg Longest Dry Spell (days/yr)
Ahmednagar	677	80.1%	75.7
Beed	721	80.5%	72.2
Latur	762	78.9%	70.4
Osmanabad	752	76.8%	71.0
Solapur	622	73.8%	76.3

High Year-to-Year Variability

Annual rainfall varies 150–185 mm around each district's mean (~20–25%), so drought must be assessed against district-specific history, not a fixed threshold.

Strong Monsoon Dependency

74–81% of annual rainfall falls in just the Jun–Sep window; the other 8 months are structurally dry, supporting season-level feature aggregation.

Clean, Near-Complete Records

9,757 daily observations per district (2000–2025) across 11 variables, with under 1% missing values overall — ready for direct use with minimal imputation.

Temperature Profile

Daily mean temperature is fairly uniform across districts (25.7–26.8°C), with summer highs near 44–45°C and winter lows of 7–9°C — informing heat-stress risk features.

Rainfall Is Trending Upward

Comparing the first and latest decade, annual rainfall rose 2–16% in every district — sharpest in Solapur (+16%), mildest in Latur (+2%).

Extreme Dry Spells

The longest recorded dry spell reaches 125–133 days in Solapur and Beed — over 4 months with no meaningful rain — a worst-case benchmark for contingency planning.

Crop Data — Yield Trends & District Gaps

District Economic Survey records (2000–01 to 2022–23) across 7 crop-season combinations

Crop-Season	Mean Area (Hectare)	Mean Yield (Tonne/Hectare)	Missing Data
Arhar/Tur (Kharif)	54,261	0.61	0%
Bajra (Kharif)	71,562	0.53	0.9%
Jowar (Kharif)	40,904	0.78	7.0%
Jowar (Rabi)	305,949	0.64	0%
Soyabean (Kharif)	109,932	1.07	0%
Sunflower (Kharif)	8,336	0.46	0.9%
Sunflower (Rabi)	14,320	0.55	6.1%

Jowar Anchors the Rabi Season

Jowar (Rabi) covers ~306k hectares on average — by far the largest crop-season by area — while Soyabean (Kharif) delivers the highest mean yield (1.07 t/ha).

Large District-Level Yield Gaps

Jowar (Kharif) yield ranges from 1.10 t/ha in Latur to 0.56 t/ha in Solapur (up to 53% variability) — the same district gap seen in the rainfall data.

Mostly Complete, Some Gaps

115 district-year records (5 districts × 23 years) with 0–7% missing values, concentrated in Jowar (Kharif) and Sunflower (Rabi) — flagged for imputation before modeling.

System Architecture — Pipeline Overview

Drought-aware crop-productivity decision support for 5 Maharashtra districts — 10 stages, grouped into 4 phases

DATA SOURCES

Daily Weather
5 districts · 2000–2025 · 11 variables/day

DES Crop Data
Area / Production / Yield by district, crop, season

Contingency Plans
5 official district PDF

Census 2011
Population, density, workforce

I Weather Forecasting

Stages 1–2

1 Select the target district and crop-season (Kharif / Rabi)

2 Per-district model forecasts the next 3 days of 11 weather variables from a rolling lookback window — no live weather API needed

II Condition & Yield

Stages 3–6

3 Aggregate daily weather into season rainfall, longest dry spell & monsoon onset

4 Classify as Normal / Dry spell / Deficit / Drought-like vs. the district's own history

5 Predict yield with Random Forest / Gradient Boosting per district-crop-season

6 Explain each prediction with SHAP driver scores

III Impact & Comparison

Stages 7–8

7 Convert the predicted yield gap into human impact using Census population & workforce

8 Normalise severity & exposure scores to rank all 5 districts side by side

IV Knowledge Graph & LLM

Stages 9–10

9 Query the knowledge graph built from the 5 contingency plans for matching official actions

10 LLM combines model outputs + retrieved facts into one grounded advisory answer

Data Preparation & Predictive Modeling

The plan for turning four raw sources into one clean modeling table, and the modeling approach for stages 3–8

A. Data Preparation Plan

1 Re-parse the HTML-mislabeled ".xls" files (DES crop table, both Census tables) into clean CSVs.

2 Standardize district names to one canonical key across DES, Census, contingency-plan and weather sources.

3 Standardize crop names to one canonical key (e.g. "Bajra" vs. "Pearl millet") across DES and the plans.

4 Parse DES agri-year strings ("2000–2001") into an integer crop year for joining.

5 Extract each district's static profile — rainfall normal, soil %, land use, irrigation % — from contingency-plan pages 1–2.

6 Extract the condition→action tables from all 5 plans into one structured lookup — this fixes the vocabulary the classifier and knowledge graph both use.

7 Aggregate daily weather into one season-level row per district-year-season, reused by every crop grown that season.

B. Predictive Modeling Approach (Stages 3–8)

3 Season Aggregation

Daily weather is summarised into seasonal rainfall, longest dry spell and monsoon onset for each district and season.

4 Condition Classification

Each season is labelled using rules based on the district's own rainfall history, since no labelled drought data exists.

5 Yield Prediction

Yield is predicted using Random Forest and Gradient Boosting, as the data is too small for deep learning.

6 Explainability

SHAP explains how much each factor, such as rainfall or dry spell length, affected the predicted yield.

7 Human-Impact Translation

The predicted yield loss is combined with Census data to estimate how many people are affected.

8 Cross-District Comparison

Severity and exposure scores are normalised and combined to rank all five districts by priority.

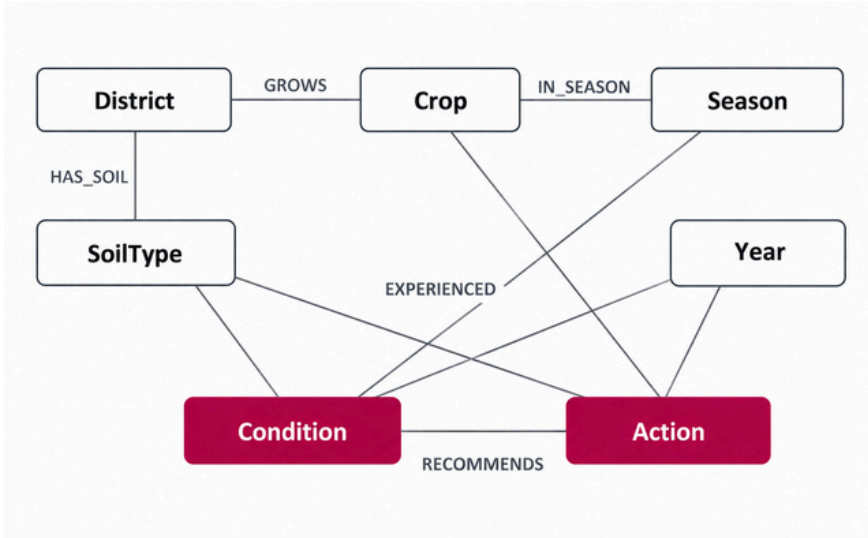
Knowledge Graph, GraphRAG & LLM Layer

How stages 9–10 turn model outputs and official contingency plans into one grounded, natural-language answer

Why a Knowledge Graph

Districts share soil types, crops span multiple districts and seasons, and conditions map to actions that depend on crop and soil type together. A knowledge graph answers multi-hop questions a flat lookup table can't, and needs no data beyond what data preparation already produces.

Nodes - District, Crop, Season, SoilType, Condition, Action, Year
Key edges - HAS_SOIL, GROWS, IN_SEASON, EXPERIENCED, RECOMMENDS
Scale - 5 districts, up to 7 crops, a handful of conditions and soil types — small enough for Neo4j or a local NetworkX graph.



GraphRAG Pipeline

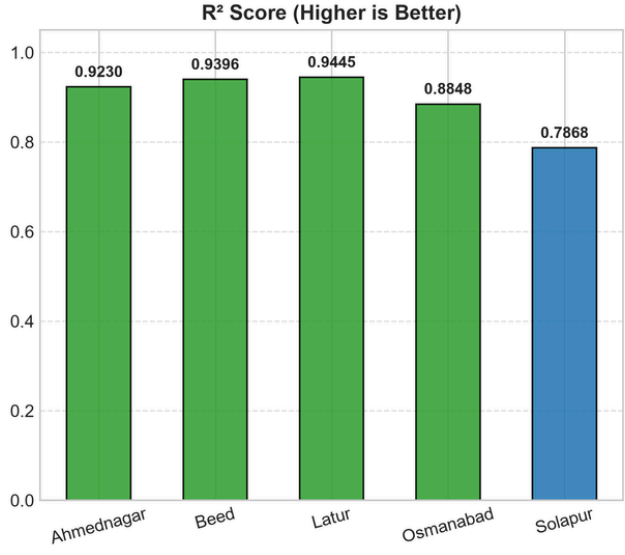
1. Natural-language question
2. LLM extracts entities — district, crop, condition
3. Query retrieves the matching subgraph
4. Retrieved facts combined with stage 5–8 model outputs
5. LLM writes one grounded natural-language answer

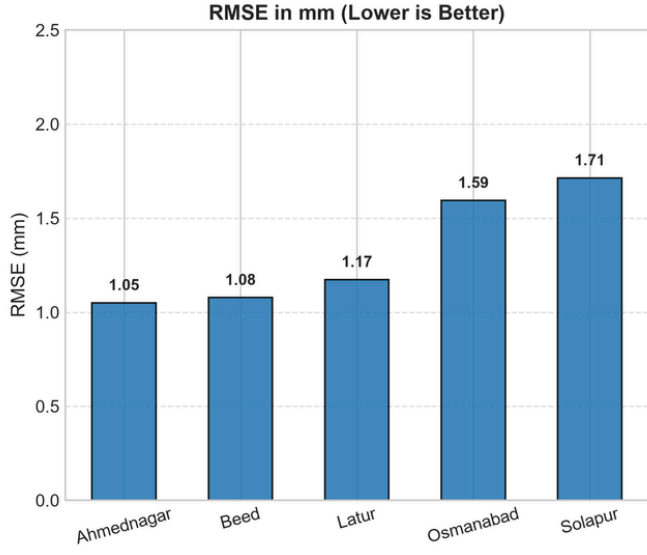
! **Non-negotiable guardrail.** The LLM's only inputs are facts retrieved from the graph and numeric outputs already produced by stages 5–9 — it never predicts a yield number or invents a recommendation on its own.

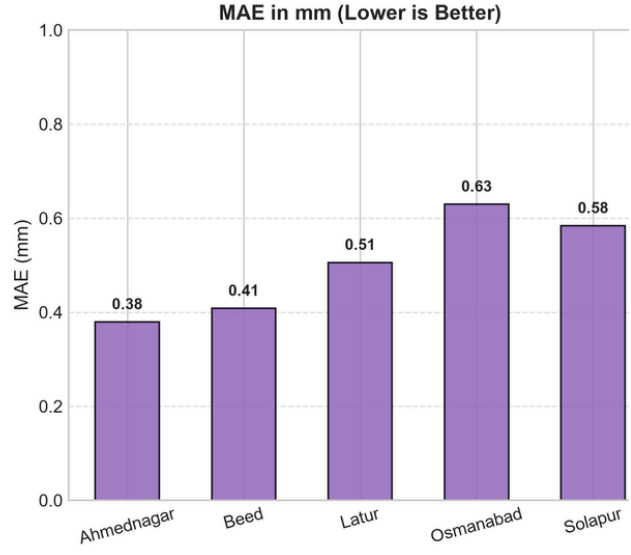
Results – Quantitative Performance

District\Model	SARIMA			XGBoost			LSTM		Hurdle			TFT		
	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2	RMSE	R^2	RMSE	MAE	R^2	RMSE	MAE
Ahmednagar	0.432	0.152	0.118	0.3025	6.2105	1.4178	0.7485	3.9198	***0.923***	***1.0495***	***0.3792***	0.5105	2.7707	1.4311
Beed	0.479	0.174	0.136	0.506	4.0624	1.4435	0.7504	3.7532	***0.9396***	***1.0774***	***0.408***	0.4386	3.3375	1.5963
Latur	0.592	0.891	0.672	0.5902	3.6407	1.6473	0.7377	3.8272	***0.9445***	***1.1732***	***0.5057***	0.4995	3.8175	1.8014
Osmanabad	0.544	1.07	0.814	0.5486	4.0187	1.5234	0.7246	3.8766	***0.8848***	***1.5937***	***0.6296***	0.4389	4.0426	1.8354
Solapur	0.512	3.62	2.741	0.509	3.4419	1.4598	0.7358	3.7281	***0.7868***	***1.7136***	***0.5837***	0.3715	3.2945	1.4836

Final Per-District Hurdle Ensemble Model Performance Metrics







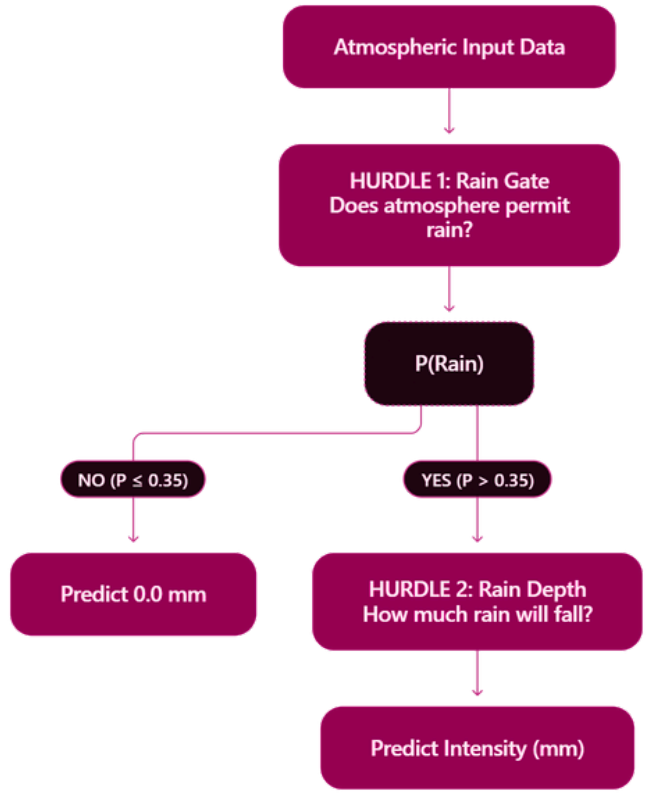
Results Analysis & Interpretation

Why Standard Models Fall Short — The Hurdle Advantage

Standard Models	Hurdle Model
• Treat rainfall as a single continuous prediction	• Splits rainfall into occurrence + intensity
• >80% dry days dominate the target distribution	• Classifier explicitly identifies rain vs. no rain
• Small false-positive rainfall values accumulate error	• Dry-day predictions are gated to 0 mm
• Skewed rainfall makes extreme events difficult to learn	• Intensity model trains only on rainy days

Hurdle Model

- Splits rainfall prediction into two stages — whether it will rain and how much it will rain.
- Hurdle 1 – Rain Gate: A classifier checks the probability of rain; P ≤ 0.35 → 0 mm, while P > 0.35 → proceed to intensity prediction.
- Hurdle 2 – Rain Depth: A regressor predicts the rainfall intensity only when rain is likely, using log(1 + rainfall) to handle highly skewed rainfall values.



Results Analysis & Interpretation

Model	Why it struggled / performed differently
SARIMA	• Assumes mainly linear temporal patterns.
	• Struggles with nonlinear and highly variable rainfall patterns.
XGBoost	• Doesn't explicitly separate rain occurrence and intensity.
	• Extreme rainfall can increase prediction error.
LSTM	• Rainfall is intermittent and noisy.
	• Doesn't explicitly distinguish dry and rainy regimes.
TFT	• Directly predicts rainfall as a continuous value.
	• Doesn't explicitly handle zero-inflated rainfall.
Hurdle	• Separates rain/no-rain prediction.
	• Separately models rainfall intensity.

REMAINING WORK & COMPLETION PLAN

Task Sequence & Status

3 of 7 tasks complete — remaining work follows the pipeline stage order

1	Data preprocessing & unified modelling table	Completed
2	Exploratory data analysis	Completed
3	Weather forecasting models — LSTM, Hurdle Ensemble, XGBoost, SARIMA, TFT	Completed
4	Season aggregation & condition classification	Pending
5	Yield prediction, SHAP explainability & district impact ranking	Pending
6	Knowledge graph + GraphRAG / LLM advisory layer	Pending
7	MLOps, integration, testing & final report	Pending

TEAM CONTRIBUTION & RESPONSIBILITY

Team Roles Across the 10-Stage Pipeline

Equal ownership — each member owns one model or component already built plus an equal share of the remaining stages

V Vyshnav Kumar S

Weather Forecasting & Knowledge Graph

- Trained per-district PyTorch LSTM weather forecaster ✓
- Feed model forecasts into the season-aggregation stage
- Extract (condition, action) tables from contingency plans
- Build the knowledge graph — schema + Neo4j

G Gowtham S D

Hurdle Ensemble & GraphRAG / LLM

- Trained the TFT and Two-Stage Hurdle Ensemble pipeline for weather forecasting ✓
- GraphRAG entity extraction — district, crop, condition
- Templated Cypher retrieval of matching subgraphs
- Guardrailed LLM advisory answer

G Guhan K B

Yield Prediction & Forecast Refinement

- Trained per-district XGBoost model for weather forecasting ✓
- Season aggregation & condition classification
- Yield model — RF / Gradient Boosting + linear baseline
- MLOps: tuning & scheduled retraining of forecast models

V Vaishnavi Gupta

Explainability, Impact & MLOps

- Trained per-district SARIMA model for weather forecasting ✓
- SHAP explanations of yield drivers
- Census human-impact & cross-district ranking
- MLOps: automated data ingestion, versioning & monitoring

+ Shared by all four. Data preprocessing, EDA and model benchmarking (completed) · system integration, end-to-end testing, final report and paper writing (upcoming).

References

[1] Mulla, S., Pande, C.B. & Singh, S.K. (2024). "Time Series Forecasting of Monthly Rainfall using Seasonal Auto Regressive Integrated Moving Average with Exogenous Variables (SARIMAX) Model." Water Resources Management, 38, 1825-1846.
[2] S. N. H. Bukhari and K. A. Ogudo, "Forecasting temperature and rainfall using deep learning for the challenging climates of Northern India," PeerJ Computer Science, vol. 11, p. e3012, 2025, doi: 10.7717/peerj-cs.3012.
[3] Bryan Lima, Sercan Ö.Arıkb, Nicolas Loeffb, Tomas Pfisterdoi : Temporal Fusion Transformers for interpretable multi-horizon time series forecasting 10.1016/j.ijforecast.2021.03.012
[4] Dehvari, Masoud, Saeed Farzaneh, and Ehsan Forootan. "Forecasting rainfall events based on zenith wet delay time series utilizing eXtreme gradient boosting (XGBoost)." Advances in Space Research 75, no. 3 (2025): 2584-2598.

THANK YOU